Semi-symmetric fold sequences: what is known

Summary of the `mhuang03/folds` project

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Abstract

A strip of paper is divided into N panels, and the panels are bisected one at a time. A bisection order is *semi-symmetric* if, at every step, the bisected panel sees equally many earlier creases on its two sides within its folding window. We study the number $a(N)$ of such orders. We find no closed form or recurrence in previous terms, and we prove that the ordinary generating function is not algebraic. We do obtain: several equivalent reformulations (including a doubling-map form); an exact subset recursion; a congruence $a(N) \equiv c(N) \pmod 4$; a classification of the "neutral" splittings with an exact decomposition identity; a generator for valid orders of any length (truncated bit reversal); a complete description of which two-step openings can be completed; and super-exponential growth bounds $(N/4096)^{N/16} \le a(N) \le N!\, e^{-0.0878N(1+o(1))}$. Results marked [Lean] are machine-checked in Lean 4 with Mathlib, with no `sorry` and no `native_decide`.

1 The problem

Number the panels $1, \dots, N$. A *fold sequence* is a permutation $a = (a_1, \dots, a_N)$ of $[N] = \{1, \dots, N\}$. At step k the panel $m = a_k$ is bisected by folding its nearer edge onto a reference crease; the panels caught between are $m \pm d$ for $1 \le d \le r$, where

$$r = r(m) = \min(m - 1,\, N - m).$$

Let $S = \{a_1, \dots, a_{k-1}\}$ be the set of panels bisected before step k. Define the *left arm* $L(m) = \{m - r, \dots, m - 1\}$ and the *right arm* $R(m) = \{m + 1, \dots, m + r\}$.

Definition 1. The step (S, m) is *symmetric* if $m - d \in S \iff m + d \in S$ for every $1 \le d \le r$, and *semi-symmetric* if $|S \cap L(m)| = |S \cap R(m)|$. A fold sequence is (semi-)symmetric if every step is. Let $a(N)$ be the number of semi-symmetric sequences of $[N]$, and let $\theta(N)$ be the number of symmetric ones.

The window depends on the panel $m = a_k$, not on the time k. Symmetric implies semi-symmetric. The values of $a(N)$ begin:

> 1, 2, 4, 10, 28, 96, 256, 1120, 4256, 17878, 70960, 383476, 1620328, 9167908, 47821296, 276182312, 1481994716, 10084180418, 55082188120, 398874108930, … $\quad (N = 1, \dots, 20)$.

The terms are known exactly up to $N = 38$, for example

$$a(38) = 158849542048167776206921874682.$$

They were computed with a meet-in-the-middle subset dynamic program (Theorem 6), $N = 38$ took 4.6 hours. The sequence is not in the OEIS.

2 Equivalent formulations

Proposition 2 (symmetric = 3-AP-free)**.** *A sequence is symmetric if and only if there are no $x < y < z$ with $x + z = 2y$ such that y is bisected between x and z (in time). Hence $\theta(N)$ is OEIS A003407. [Lean: `symmetric_iff_apFree`]*

Proof. A symmetric step fails at m exactly when, for some $d \le r(m)$, one of $m \pm d$ comes before m and the other comes after it. Every 3-term progression $(m-d, m, m+d)$ inside $[N]$ has $d \le r(m)$. □

Write $P(j) = |S \cap [1, j]|$ and $s = |S|$. Extend P periodically by $\tilde{P}(j) = P(j \bmod N) + \lfloor j/N \rfloor s$.

Proposition 3 (doubling form)**.** *The step (S, m) is semi-symmetric if and only if $\tilde{P}(2m-1) = 2P(m-1)$. [Lean: `semiStep_iff_cyclicStep`]*

Proof. Suppose $2m - 1 \le N$, so $r = m - 1$. The left arm count is $P(m-1)$. Since $m \notin S$, the right arm count is $P(2m-1) - P(m) = P(2m-1) - P(m-1)$. Equating the two gives $P(2m-1) = 2P(m-1)$.

Otherwise $r = N - m$. The left arm count is $P(m-1) - P(2m-1-N)$ and the right arm count is $s - P(m-1)$. Equating them gives $s + P(2m-1-N) = 2P(m-1)$, and the left-hand side is $\tilde{P}(2m-1)$. □

In terms of the discrepancy $E(x) = P(x) - sx/N$ on $\mathbb{R}/N\mathbb{Z}$, the condition reads $E(2c) = 2E(c)$ at the panel centre $c = m - \frac{1}{2}$. So the problem is governed by the doubling map $x \mapsto 2x \bmod N$. This explains several recurring features below: powers of 2, parity classes, and the special role of $N/3$ and $2N/3$, which form a 2-cycle of the doubling map.

Proposition 4 (window rule)**.** *Let B and A be the numbers of elements of S on the near side and the far side of m. The* near side *is the one towards the closer end of the strip; when $m = (N+1)/2$, call either side near. Then (S, m) is semi-symmetric if and only if $B \le A$ and exactly B elements of S on the far side lie within distance r of m. [Lean: `WindowRule`]*

Proof. The near arm reaches the end of the strip, so it contains all B near-side elements. The far arm contains the far-side elements within distance r, and there are at most A of them. □

Proposition 5 (reversal and mirror)**.** *If a is semi-symmetric, then so are its reversal* $\mathrm{rev}(a) = (a_N, \ldots, a_1)$ *and its mirror* $\mathrm{mir}(a) = (N+1-a_1, \ldots, N+1-a_N)$*. [Lean: `Basic`]*

Proof. The mirror case is clear. For the reversal, consider panel m placed at step k. In $\mathrm{rev}(a)$, the panels placed before m are exactly the panels placed after m in a. Inside the window these are $(L \cup R) \setminus S$, where S is the earlier set in a. Since $|L| = |R| = r$, we have $|L \setminus S| = |R \setminus S|$ if and only if $|L \cap S| = |R \cap S|$. □

3 Exact counting

For $X \subseteq [N]$, let $g(X)$ be the number of orderings of X that are semi-symmetric as prefixes of a fold sequence of $[N]$. So $g(\emptyset) = 1$ and $a(N) = g([N])$.

Theorem 6 (subset recursion)**.** *For $X \neq \emptyset$,*

$$g(X) = \sum_{\substack{m \in X \\ (X \setminus \{m\},\, m)\ \mathit{valid}}} g(X \setminus \{m\}).$$

Moreover, for every $0 \le k \le N$,

$$a(N) = \sum_{|X|=k} g(X)\, g([N] \setminus X).$$

[Lean: `g_eq_sum, semiCount_eq_sum_split`]

The second identity uses Proposition 5: the reversed suffix of a valid sequence is a valid ordering of the complement. This identity is the basis of the fast counter. It runs over orbits of the mirror, with $k = N/2$, and uses CRT residues.

Theorem 7 (a single constraint)**.** *For a fixed panel* m *with* $r = r(m)$*, the number of permutations of* $[N]$ *whose step at* m *is semi-symmetric (all other steps unconstrained) is*

$$N! \cdot \frac{4^r\, r!^2}{(2r+1)!}.$$

[Lean: `Wallis`]

Proof. Only the relative order of the $2r+1$ panels in $\{m\} \cup L \cup R$ matters. The step is valid exactly when, for some j, j panels of each arm precede m. The fraction of valid relative orders is

$$\frac{1}{(2r+1)!} \sum_{j=0}^{r} \binom{r}{j}^2 (2j)!\,(2r-2j)! = \frac{r!^2}{(2r+1)!} \sum_{j=0}^{r} \binom{2j}{j}\binom{2r-2j}{r-j} = \frac{4^r\, r!^2}{(2r+1)!},$$

by the classical identity $\sum_j \binom{2j}{j}\binom{2r-2j}{r-j} = 4^r$. □

Treating the N constraints as independent gives the *Wallis model* $a_W(N) = N! \prod_m \frac{4^{r(m)} r(m)!^2}{(2r(m)+1)!}$. The ratio a/a_W stays within 13% for $N \le 11$, but it grows like $e^{0.09N}$ in the data. So the constraints are mildly positively correlated.

Prefix counts. Let $T(N,k)$ be the number of valid k-step prefixes. For each fixed k, $T(N,k)$ is a quasi-polynomial in N of degree k: it counts lattice points in a union of rational polytopes, one per order type. Two cases are proven in Lean:

$$T(N,2) = \left\lfloor \tfrac{N^2}{2} \right\rfloor, \qquad 36\,T(N,3) = 8N^3 - 3N^2 - \begin{cases} 30N & N \text{ even} \\ 12N & N \text{ odd} \end{cases} + (0,7,8,-9,16,-1)_{N \bmod 6}.$$

[Lean: `WindowRule`]

Exact formulas are also known for $k \le 6$. The periods follow a *period law*, proven on paper: the period of $T(N,k)$ is $B(k)$ for every $k \ge 4$, where

$$B(k) = \operatorname{lcm}\big(2^{k-1},\ 2^a(2^s-1) : a+s = k-1\big),$$

giving $1, 1, 2, 12, 168, 1680, 104160, 624960$. The factors 2^a come from halving chains and the factors $2^s - 1$ from periodic cycles of the doubling map, as Proposition 3 predicts. The bivariate generating function $\sum T(N,k)\, t^N u^k$ is not D-finite.

A congruence. Let $c(N)$ be the number of valid sequences fixed by $\mathrm{rev} \circ \mathrm{mir}$.

Theorem 8. *For $N \geq 2$, $a(N) \equiv c(N) \pmod 4$. Also, $c(N) = 0$ when $N \equiv 3 \pmod 4$, so $4 \mid a(N)$ for those N. [Lean: `semiCount_mod_four`]*

Proof. By Proposition 5, the Klein group $\{1, \mathrm{rev}, \mathrm{mir}, \mathrm{rev} \circ \mathrm{mir}\}$ acts on valid sequences. For $N \geq 2$, neither rev nor mir fixes a permutation. Burnside's lemma then says that $(a + 0 + 0 + c)/4$ is an integer, so $a \equiv -c \pmod 4$. The reversal acts on the fixed set of $\mathrm{rev} \circ \mathrm{mir}$ without fixed points, so c is even, and therefore $-c \equiv c \pmod 4$.

For odd N, the middle panel is always placed at an odd step. A sequence fixed by $\mathrm{rev} \circ \mathrm{mir}$ must place it at step $(N + 1)/2$, which is even when $N \equiv 3 \pmod 4$. □

No further modular structure was found. In particular, $a(N) \bmod 8$ is not determined by $c(N) \bmod 8$, and there is no linear recurrence for $a(N)$ modulo any prime $p \leq 97$.

4 Structure

Theorem 9 (parity blocks)**.** *Let σ be an ordering of the even panels and τ an ordering of the odd panels. The sequence "σ, then τ" is valid if and only if σ and τ are valid after the relabellings $2y \mapsto y$ and $2y - 1 \mapsto y$, respectively. The same holds with the odd panels first. Consequently $a(2n) \geq 2a(n)^2$, $a(2n + 1) \geq 2a(n)a(n + 1)$ and $a(N) \geq 2^{N-1}$. [Lean: `semiSymmetric_blocks_iff`, `Counting`]*

More generally, every residue class modulo q is a *window copy*: it behaves exactly like a smaller strip. This gives a *composition* inequality $a(qn) \geq a(q)\, a(n)^q$, with a version for every $N = nq + \rho$. [Lean: `Growth`, `SuperExp6`]

Neutral sets. Call $X \subseteq [N]$ *neutral* if $|X \cap L(m)| = |X \cap R(m)|$ for every $m \in [N]$. If a prefix set is neutral, the sequence splits into two independent problems. By Proposition 3, X is neutral if and only if $P(2k+1) = P(k) + P(k+1)$ for all k. From this one shows $|N \cdot P(t) - |X|\, t| \leq \min(|X|, N - |X|)$: neutral sets are balanced in the Sturmian sense.

Theorem 10 (classification of neutral sets)**.** *For $N \geq 2$, the nonempty proper neutral subsets of $[N]$ are exactly:*

- *the odd panels and the even panels;*
- *$E_d = \{x \in [N] : x \bmod d \in \{2, 4, \ldots, d - 1\}\}$ and its complement, for each odd divisor $3 \leq d < N$ of N.*

They form an antichain. Consequently, with $j = (d - 1)/2$,

$$a(N) = 2\, a(\lfloor \tfrac{N}{2} \rfloor)\, a(\lceil \tfrac{N}{2} \rceil) + 2 \sum_{\substack{d \mid N,\ d \text{ odd} \\ 3 \leq d < N}} a(\tfrac{jN}{d})\, a(\tfrac{(j+1)N}{d}) + I(N),$$

where $I(N)$ counts the valid sequences that have no neutral proper prefix set. [Lean: `neutral_classification`, `identity_A`]

Numerically $I(N)/a(N) \to 1$ (0.999 at $N = 24$), so this identity is exact but does not by itself give a recursion.

A generator for every N. Choosing, at every node of the binary residue tree (classes modulo 2^d), which half goes first gives a family of exactly 2^{N-1} sequences, all of them symmetric. [Lean: `gen_valid, card_parFam`]

Theorem 11 (truncation)**.** *Deleting the panels larger than N from a member of the family for $M \geq N$ gives the corresponding member for N. In particular, for any $2^k \geq N$, the bit-reversal order of $\{1, \ldots, 2^k\}$ with the panels $> N$ removed is symmetric and semi-symmetric. Its i-th term can be computed in $O(\log N)$ time. [Lean: `truncation, gen_filter, genAt_eq`]*

For example, for $N = 6$ and $k = 3$, the bit-reversal order $1, 5, 3, 7, 2, 6, 4, 8$ truncates to the valid sequence $1, 5, 3, 2, 6, 4$.

Which openings survive. Some valid prefixes cannot be completed. For example, at $N = 4$ the opening $1, 4$ leaves no valid move.

Theorem 12 (two-step liveness)**.** *Let $[m_1, m_2]$ be a valid two-step prefix. It extends to a full semi-symmetric sequence unless $\{m_1, m_2\} = \{1, N\}$ and either $N \geq 4$ is even or $N = 7$. [Lean: `ThinCover.conjectureC`]*

The proof combines explicit constructions with a covering argument: window copies, Beatty sets $\{\lfloor kN/p \rfloor\}$, residue chains, and a greedy-then-reverse criterion. The remaining pairs, for all $N \geq 1000$, are covered by LP-certified polynomial inequalities, and those with $N < 1000$ by kernel computations. For three-step prefixes, the analogous classification is proven for $N \leq 50$ and checked to $N = 63$.

5 Growth

Theorem 13 (slip recursion)**.** *For every $m \geq 1$, $a(4m) \geq 2^m a(m)^4$. Consequently $a(N) \geq N^{N/8}$ for $N = 4^j$. [Lean: `semiCount_four_mul, pow_le_semiCount_pow_four`]*

Idea. Start from the parity-block sequences of $[2n]$ built from odds-first orders of $[n]$. A *slip* at k swaps the panels $2k$ and $2k + 1$ between the two parity classes. In the doubling form, each slip changes the induced conditions on the two blocks in exactly one place, and odds-first block orders satisfy the changed condition automatically when k is odd. So each of the $\approx n/2$ odd positions can be slipped independently, giving a factor $2^{n/2}$.

For the consequence, set $b_j = \ln a(4^j)$. Then $b_{j+1} \geq 4^j \ln 2 + 4b_j$, so $b_j \geq j4^{j-1} \ln 2 = \frac{1}{8} N \ln N$. □

Theorem 14 (bounds for all N)**.** *For every $N \geq 1$,*

$$\Big(\frac{N}{4096}\Big)^{N/16} \;\leq\; a(N) \;\leq\; N!\, e^{-0.0878\, N(1+o(1))},$$

and the upper bound improves to $e^{-0.1255N(1+o(1))}$ for even N. [Lean: `self_pow_le_semiCount_pow_16, MaxMoves2`]

The upper bound counts valid moves. When $|S| \equiv 2 \pmod 4$, at most $\lfloor 2N/3 \rfloor$ panels are valid next moves; a 2-adic refinement covers the other sizes of S. [Lean: `card_moves_two_mod_four, card_moves_two_adic`] The lower bound for all N uses an odd-N variant of the slips.

To raise the exponent to $1/12$ it would suffice to prove a combinatorial "Hypothesis H" about residue chains for residues $\rho \leq q/16$ [Lean: `allN_of_smallH`]. The obvious palindromic constructions provably cannot reach residues $\rho \equiv 3 \pmod 4$ when q is odd [Lean: `no_palin_three`].

Conjecture 15 (asymptotics)**.** $a(N) = N!/(\kappa_N \ln N)^N$ with κ_N bounded, so $\ln a(N) = N \ln N - N \ln \ln N + O(N)$.

The evidence is sequential Monte Carlo estimates of $\ln a(N)$ up to $N = 4096$, and a continuum limit of the prefix polytopes: the acceptance ratio decays like $0.75/(\ln k + 0.25)$.

6 Why there is (probably) no formula

Proposition 16. *$\sum_N a(N)x^N$ is not algebraic, not rational, and not C-finite.*

Proof. The coefficients of an algebraic power series grow at most exponentially, but $a(N) \geq (N/4096)^{N/16}$ grows faster than any exponential. □

If the conjecture holds, then $a(N)$ is not P-recursive either. Perron–Kreuser theory forces a holonomic sequence with $\ln a_n \sim n \ln n$ to satisfy $\limsup(a_n/n!)^{1/n} > 0$, but under the conjecture $(a(N)/N!)^{1/N} \sim 1/(\kappa \ln N) \to 0$.

The following searches were all negative, each with at least five spare equations:

- linear recurrences with polynomial coefficients, up to 30 unknowns (lattice reduction up to 64);
- algebraic and differentially algebraic generating functions (ordinary and exponential);
- Mahler and q-difference equations;
- Somos- and Riccati-type relations;
- 2-regular and automatic structure;
- Hankel determinants;
- OEIS transforms;
- about 10^6 expression trees in earlier terms.

Even the simpler symmetric count $\theta(N)$ (A003407) has no known formula. We computed it exactly up to $N = 2000$.

An arithmetic effect. Let $y(N) = \ln(a(N)/a_W(N))$. The local bump $y(N) - \frac{1}{2}(y(N-2) + y(N+2))$ is positive for every $N \equiv 0 \pmod 3$ in $6 \leq N \leq 30$, and negative for every other N in $7 \leq N \leq 31$. A plausible mechanism is the 2-cycle $\{N/3, 2N/3\}$ of the doubling map, which consists of panel boundaries exactly when $3 \mid N$.

7 Open problems

1. Prove $\ln a(N) \sim N \ln N$, or even $\liminf \ln a(N)/(N \ln N) > 1/16$.
2. Prove $a(N) \leq N!\, e^{-\omega(N)}$. This reduces to a "Path Lemma" that has been checked numerically.
3. Classify the live k-step prefixes for $k \geq 3$ and all N.
4. Explain the mod-3 effect.

Sources. The Lean files are in `Folds/`; the fast core library builds in a few minutes. The paper proofs and data are in `experiments/*/PROOFS.txt`, and the exact counter is in `rust/`.